MIL-L-11931C
2 September 1983
SUPERSEDING
MIL-L-11931B
30 April 1973

MILITARY SPECIFICATION

LENS, CAMERA COPYING LE-5( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a still picture camera lens and is designated Lens, Camera Copying LE-5( ) (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-C-675	- Coating Of Glass Optical Elements (Anti-reflection).
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronics Equipment And Parts, Packaging Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-130	- Identification Marking Of US Military Property.
MIL-STD-150	- Photographic Lenses.
MIL-STD-252	- Classification Of Visual And Mechanical Defects For Equipment, Electronic, Wired, And Other Devices.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and Standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Lens, Camera Copying LE-5( ) is a lensboard-mounted, nominal 12 inch (30.48cm) focal length lens in a between-the-lens shutter, with a front lens cap and a cable release.

3.2 First article. When specified, samples shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Parts, materials, and processes. Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements as specified herein.

3.3.1 Recycled or reclaimed materials. Recycled or reclaimed materials in lieu of virgin materials shall be used to the maximum extent possible without jeopardizing the specified performance and reliability of the lens.

3.4 Construction. Lens, Camera Copying LE-5( ) shall have the properties and requirements as defined in MIL-STD-150 and specified herein.

3.4.1 Lens type. The lens shall be a type XII, semi-apochromatic copy lens.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the current version before use.

3.4.1.1 Focal length. The equivalent focal length shall be 12 ±.48 inches (30.48 ±1.22cm).

3.4.1.2 Relative aperture. The maximum relative aperture shall be f/8, or faster.

3.4.1.3 Field of view. The field of view at maximum relative aperture shall be 8 by 10 inches (20.32 by 25.4cm).

3.4.1.4 Resolving power. When tested as specified in 4.7.4, the photographic resolving power of the lens shall be not less than 14 lines per millimeter at any point of the 8 by 10 inch (20.32 by 25.4cm) field of view.

3.4.1.5 Distortion. There shall be no tangential distortion. Radial distortion shall not exceed 1.5mm.

3.4.1.6 Reflection-reducing coating. All air-glass surfaces shall be reflection-reducing coated in accordance with MIL-C-675.

3.4.1.7 Iris diaphragm. The lens shall be equipped with a continuously variable iris diaphragm and an iris diaphragm control. The iris diaphragm control shall be marked with the following minimum range of indices: 8, 11, 16, 22, 32 and 45; each representing the f-number of the effective aperture associated with the position of the iris diaphragm control against a fixed index mark for that particular number.

3.4.1.8 Lens mounting. The lens shall be fitted into a between-the-lens shutter and attached to a square lensboard by means of a separable retaining flange. The lensboard, which shall be of the insert type, shall be of mahogany having a smooth, rubbed finish, and shall be 11 ±.031 inches (27.94 ±.079cm) long by 11 ±.031 inches (27.94 ±.079cm) wide by 0.469 ±.016 inch (1.19 ±.041cm) thick. A rabbetted edge shall be provided to fit the opening in the front of the camera.

3.4.2 Shutter. The lens shall be fitted into a heavy duty, set and release type, between-the-lens shutter, Ilex No. 4 Acme Synchro, or equal. The shutter shall incorporate the following features:

a. A minimum range of marked shutter speeds in either of the two following sequences:

(1) 1, 1/2, 1/5, 1/10, 1/25, 1/50, 1/100, 1/150 second, B(bulb), and T(time).

(2) 1, 1/2, 1/4, 1/8, 1/15, 1/30, 1/60, 1/125 second, B(bulb), and T(time).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

b. Built-in synchronization for electronic flash and for class M flash lamps.

c. Press-focus control (blade arrestor).

d. Cable release socket.

3.4.2.1 Effective exposure times. When tested as specified in 4.8, the effective exposure time (the product of total open time and efficiency) for each "instantaneous" shutter speed of one second or less shall vary from the marked speed by not more than 25 percent.

3.4.3 Cable release. A nominal 40 inches (101.6cm) long, metal or vinyl-sheathed cable release, threaded to fit the shutter's cable release socket, shall be supplied.

3.4.4 Lens cap. A snug-fitting lens cap for the front of the lens shall be supplied.

3.5 Storage requirements. Lens, Camera Copying LE-5( ) shall be able to ~~withstand storage at temperatures ranging from -60 (+5 or -0)°F (-51.1 (+2.8~~ or -0)°C) to 160 (+0 or -5)°F (71.1 (+0 or -2.8)°C) without any degradation of performance or deterioration of the lens material, anti-reflection coating, shutter or accessories.

3.6 Finishes. Exterior metal surfaces shall have a black anodized or chromium finish in accordance with MIL-F-14072. Internal, non-optical lens barrel surfaces shall be finished matte black in accordance with the applicable provisions of MIL-STD-150.

3.7 Marking. The lens shall be marked in accordance with the appropriate paragraphs of MIL-STD-150 covering the following:

a. Lens markings.

b. Cell marking.

c. Maximum aperture stated as relative aperture.

d. Focal length. Marked focal length shall be 12 inches, or either 300 or 305 millimeters.

3.8 Identification marking. In addition to the marking specified in 3.7 herein, 1 Lens, Camera Copying LE-5( ) shall bear the following identification markings, applied in accordance with MIL-STD-130:

a. Lens front.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

(1) Either the manufacturer's name, logo, trademark or brand, design name, or any combination thereof.

(2) Serial number (this number may be in the manufacturer's normal production sequence).

b. Shutter.

(1) Either the manufacturer's name, logo, trademark, brand, or model designation, or any combination thereof.

(2) "US" designating government property.

3.9 Interchangeability. Like, non-optical units, assemblies, sub-assemblies, and replaceable parts shall be in accordance with MIL-STD-454, requirement 7.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.10.1 Beauty defects. Beauty defects, while undesirable, are acceptable to the extent that their presence shall permit the equipment to meet all other requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Packaging inspection (see 4.11).

4.3 First article.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the current version before use.

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete Lens, Camera Copying LE-5( ).

4.3.2 First article inspection. The first article inspection shall consist of the inspections, specified in subsidiary documents, of the items listed in 4.4, and the inspections specified for group A, group B and group C (see tables I, II and III respectively). The inspection shall be performed in the following order: 4.4, group A, group B, and group C. After completion of group C environmental tests, conforming units shall be reinspected and shall pass group A and group B inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where required
Finish	3.6
Reflection-reducing coating	3.4.1.6

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.3.2.

4.5.1 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in table I.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and mechanical	3.10	4.6		
Iris diaphragm marking	3.4.1.7	4.6	1.0	4.0
Lens mounting	3.4.1.8	4.6	for the	for the
Shutter	3.4.2	4.6	group	group
Cable release	3.4.3	4.6		
Lens cap	3.4.4	4.6		
Equivalent focal length	3.4.1.1	4.7.1		
Relative aperture	3.4.1.2	4.7.2	1.0	
Field of view	3.4.1.3	4.7.3	for the	1/
Distortion	3.4.1.5	4.7.5	group	

1/ All defects shall be considered major.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table II and to the special inspection levels of MIL-STD-105, table I.

4.5.2.1 Group B sampling plans. The group B sampling plans, for the AQLs listed in table II, shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Resolving power	3.4.1.4	4.7.4	4.0
Effective exposure time	3.4.2.1	4.8	4.0
Interchangeability	3.9	4.10	6.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Group C sampling plans. Two sample units shall be selected at random for each group C inspection in table III from every 50 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in an order which is satisfactory to the government.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Storage temperature:		
High	3.5	4.9.1
Low	3.5	4.9.2

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified (see 6.2).

4.5.4 Reinspection of conforming group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspections may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Visual and mechanical inspection. The equipment shall be examined for the defects listed in accordance with MIL-STD-252 and table IV.

TABLE IV. Classification of visual and mechanical defects.

Classification	Defects
Major	Those defects listed in MIL-STD-150 which exceed the conditions governing beauty defects.
Minor	Those defects listed in MIL-STD-150 classified as beauty defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

4.7 Optical test procedures.

4.7.1 Equivalent focal length. The equivalent focal length of the lens shall be determined in accordance with MIL-STD-150, method 1. Nonconformance to 3.4.1.1 shall constitute failure of this test.

4.7.2 Relative aperture. The maximum relative aperture shall be determined in accordance with MIL-STD-150, method 3 or method 4. Nonconformance to 3.4.1.2 shall constitute failure of this test.

4.7.3 Field of view. The field of view shall be examined in accordance with MIL-STD-150, method 9, to determine the relative illumination. If the relative illumination of the corners of the 8 by 10 inches (20.32 by 25.4 cm) field of view at a magnification of 0.25 is less than 66 percent of axial illumination, or the resolving power of the corners of the field of view at this magnification is less than 14 lines per millimeter, the lens shall have failed the test.

4.7.4 Resolving power. The resolving power of the lens shall be determined in accordance with MIL-STD-150, method 12, using high contrast, black lines on white background targets, "white" light, a magnification of 0.5, and panchromatic microfilm developed for maximum contrast. The results shall be recorded for subsequent use in the test procedures of 4.9. Nonconformance to 3.4.1.4 shall constitute failure of this test.

4.7.5 Distortion. The lens shall be tested in accordance with MIL-STD-150, method 2. Nonconformance to 3.4.1.5 shall constitute failure of this test.

4.8 Exposure times. The shutter shall be tested as follows:

a. The total open time from the beginning of the opening period to the end of the closing period for each marked "instantaneous" shutter speed shall be measured 3 times and the averages determined. The method of measurement shall be acceptable to the government. The ambient temperature shall be 73 ± 18°F (23 ± 10°C).

b. The efficiency of the shutter for each "instantaneous" setting shall be calculated by determining the ratio of the quantity of light actually transmitted by the shutter for that particular setting to the quantity of light that would have been transmitted had the shutter been fully open for the total open time.

c. The product of the (average) total open time and the efficency shall be the effective exposure time for each marked shutter speed setting.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the currentversion before use.

d. The results shall be recorded for subsequent use in the test procedures of 4.9. Nonconformance to 3.4.2.1 shall constitute failure of this test.

4.9 Storage temperature. The test procedures of 4.9.1 and 4.9.2 shall be performed in this sequence. The pretest data shall be the results previously obtained from the procedures of 4.7.4 and 4.8. The test storage times shall be 12 hours each for the high and low temperature tests.

4.9.1 High. Lens, Camera Copying LE-5( ) shall be tested in accordance with MIL-STD-810, method 501.1, procedure 1. Step 7 operation shall consist of a repetition of the procedures of 4.7.4 and 4.8. Resultant data shall be compared with that previously obtained. Any degradation of performance shall constitute failure of this test.

4.9.2 Low. Lens, Camera Copying LE-5( ) shall be tested in accordance with MIL-STD-810, method 502.1, procedure 1. Step 7 operation shall consist of a repetition of the procedures of 4.7.4 and 4.8. Resultant data shall be compared with that previously compared. Any degradation of performance shall constitute failure of this test.

4.10 Inspection for dimensional interchangeabilty. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.9. When the listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of the retaining flange.

b. Size and form of retaining flange thread.

c. Internal dimensions of the lens cap.

d. Size and form of the cable release socket thread.

e. Size and form of the cable release thread.

f. Units, assemblies, subassemblies, and replaceable parts of the shutter.

g. Units, assemblies, subassemblies, and replaceable parts of the diaphragm assembly.

h. Internal and external dimensions of the lensboard.

4.11 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

b. Quality conformance inspection of packaging.

4.11.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.11.2 Quality conformance inspection of packaging.

4.11.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.11.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.11.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, Packing Inspection Provisions. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE V. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 3P-1-B-ØØ-00-NU-X-XX-A-ED-A.

5.1.2 Level B. 2M-1-B-ØØ-00-NU-X-XX-A-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Lens, Camera Copying LE-5( ) which is part of Camera Set, Still Picture KS-7( ), is intended for photographing flat copy, in black-and-white or color, at magnifications in the approximate range of 0.25x to 3x, on 8x10 inch (20.32 x 25.4 cm) and smaller film.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. When a first article is required for inspection and approval (see 3.2, 4.3, and 6.3).

c. Group C failure action (see 4.5.3.3).

d. Level A or B preservation and packing (see section 5).

e. When first article inspection rough handling tests are not required.

f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of units as specified in the contract. The contracting officer should include specific instructions in acquisition documents for examination, test, and approval of the first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: LE-5W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the currentversion before use.

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Air Force - 99

Preparing activity:
Army - ER
(Project 6760-0573)

Review activity:
Air Force - 70

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-11931C	LENS, CAMERA COPYING LE-5( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, TAPE ALONG THIS LINE.)

DD FORM 82 MAR 1426 EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:08Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)* mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

COMMANDER
US ARMY ELECTRONICS RESEARCH AND DEVELOPMENT COMMAND
COMBAT SURVEILLANCE AND TARGET ACQUISITION LABORATORY
ATTN: DELCS-PE
FORT MONMOUTH, NEW JERSEY 07703

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:08Z
Check the source to verify that this is the current version before use.